FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – III
PAPER –1
TEST DATE: 31-07-2022

ANSWERS, HINTS & SOLUTIONS

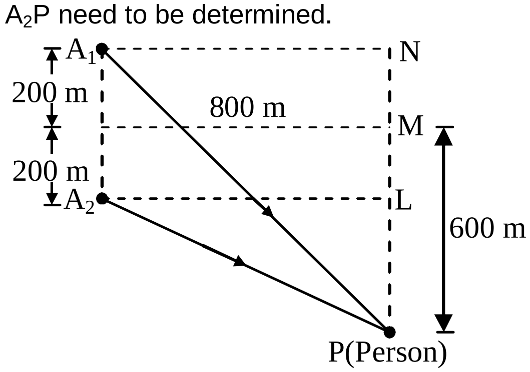
Physics

PART – I

Section – A

1. B

Sol. So as to obtain path difference between the signal from A_1 and A_2 at P, the path length A_1P and A_2P need to be determined.



In ΔA_2LP, $(A_2P)^2 = (PL)^2 + (A_2L)^2$

$\therefore$ $(A_2P)^2 = (600-200)^2 + (800)^2$

$= (400)^2 + (800)^2$

$\therefore$ $A_2P = 894.43\text{m}$

In ΔA_1NP, $(A_1P)^2 = (PN)^2 + (A_1N)^2$

$= (600+200)^2 + (800)^2$

$\therefore$ $A_1P = 1131.37\text{m}$

Path difference $\Delta = A_1P - A_2P = 236.94\text{m}$

Since the person is at the position of third order maximum,

$\therefore$ $\Delta = 3\lambda$ $\quad [\Delta = n\lambda]$

or $3\lambda = 236.94\text{m}$

$\therefore$ $\lambda \approx 79\text{m}$

2. A

Sol. If at a distance r from the centre of the earth the body has velocity v, by conservation of mechanical energy,

$$\frac{1}{2}mv^2+\left(-\frac{GMm}{r}\right)=\frac{1}{2}mv_e^2+\left(-\frac{GMm}{R}\right)$$

or $$v^2=v_e^2+\frac{2GM}{R}\left[\frac{R}{r}-1\right]$$

But as $v_e=\sqrt{2gR}$ and $g=\left(GM/R^2\right)$

$$v^2=2gR+2gR\left[\left(R/r\right)-1\right]$$

or $$v=\sqrt{\frac{2gR^2}{r}},\quad \text{i.e,}\quad \frac{dr}{dt}=R\frac{\sqrt{2g}}{\sqrt{r}}$$

i.e, $$\int_0^t dt=\frac{1}{R\sqrt{2g}}\int_R^{R+h}r^{1/2}dr$$

i.e, $$t=\frac{2}{3}\frac{1}{R\sqrt{2g}}\left[\left(R+h\right)^{3/2}-R^{3/2}\right]$$

i.e, $$t=\frac{1}{3}\sqrt{\frac{2R}{g}}\left[\left(1+\frac{h}{R}\right)^{3/2}-1\right]$$

3. D

Sol. Initial activity if nuclei $A_0=\lambda N_0$

Activity if nuclei at time t, $A=\lambda N$

$$=\lambda N_0e^{-\lambda t}$$

$$A=A_0e^{-\lambda t} \quad \text{....(i)}$$

Since activity decreases at 4.0% per hour, so activity of ^{55}Co radio nuclide at $t=1$hr,

$$A=A_0-\eta A_0=A_0\left(1-\eta\right),\ \text{where}\ \eta=0.04 \quad \text{...(ii)}$$

Taking log of Eq. (i), we get

$$\log\frac{A}{A_0}=-\lambda t \quad \text{...(iii)}$$

or $$\log\frac{A_0\left(1-\eta\right)}{A_0}=-\lambda t$$

or $$\lambda=-\frac{1}{t}\ln\left(1-\eta\right)$$

On substituting values, we get

$$\lambda\simeq1.1\times10^{-5}s^{-1}$$

4. A

Sol. Amplitude of oscillation of block initially

$$A=\frac{2mg}{k}$$

Velocity when it strikes upper block

$$\upsilon=\sqrt{\frac{k}{m}\left[\left(\frac{2mg}{k}\right)^2-\left(\frac{mg}{k}\right)^2\right]}=\sqrt{3}\sqrt{\frac{k}{m}}\left(\frac{mg}{k}\right)$$

New equilibrium is at $x=\frac{mg}{2k}$ from relaxed state of upper spring.

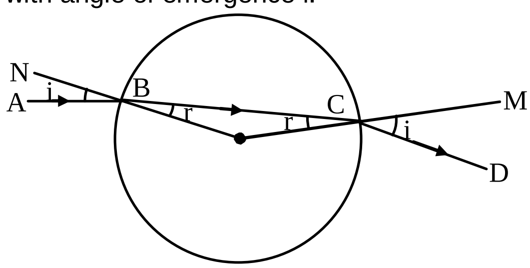
If A' is new amplitude then

$$\upsilon = \sqrt{\frac{2k}{m}}\sqrt{A'^2 - \left(\frac{mg}{2k}\right)^2} = \sqrt{3}\sqrt{\frac{k}{m}}\left(\frac{mg}{k}\right)$$

$$A' = \frac{\sqrt{7}}{2}\frac{mg}{k} = 5 \text{ cm}$$

5. BC

Sol. The diagram shows the path of a ray AB incident at B on the sphere at angle i. The path ABCD shows the refracted ray BC, inside the sphere with angle r of refraction and the emergent ray CD with angle of emergence i.



Obviously $\angle OBC = \angle OCB = r$.
Hence $\angle ABN = \angle DCM = i$ for all values of i.
Also, if $\angle ABN = 90°$, grazing incident at B, then
$\angle DCM = 90°$, grazing emergence at C.
But AB and CD will not be parallel.

6. ACD

Sol. AT 50 Hz,

$I_1 = I_2$

$X_L = X_C$

$$Z = R = \frac{V^2}{P} = 50\Omega$$

$$I_1 = I_2 = \frac{100/\sqrt{2}}{50} = \sqrt{2}\text{ A}$$

$$I_{net} = I_1 + I_2 = 2\sqrt{2}\text{ A}$$

LCR branch behaves as a pure resistive branch

7. AB

Sol. $BvI = L\frac{di}{dt}$

$\Rightarrow \quad \frac{Bl}{L} \times vdt = di$

$\Rightarrow \quad \frac{Bl}{L}\int_0^x dx = \int_0^i di$

$\Rightarrow \quad \frac{Blx}{L} = i \qquad$ ....(i)

Now $\quad F = ma = mg - Bil \qquad$ ....(ii)

From eqns. (i) and (ii)

$$\frac{d^2x}{dt^2} = g - \frac{B^2l^2}{mL}x$$

$\Rightarrow \qquad \omega^2 = \frac{B^2l^2}{mL} = 1$ and amplitude $A = g \times \frac{mL}{B^2l^2}$

8. ACD

Sol. $U = \frac{1}{2}\varepsilon_0 E^2 \times \text{volume}$

$$= \frac{1}{2}\varepsilon_0 \left(\frac{Q_0}{\ell^2 \varepsilon_0}\right)^2 (\ell - x)\ell d$$

$$= \frac{1}{2}\frac{Q_0^2}{\ell^3 \varepsilon_0}(\ell - x)d$$

$$F = \frac{-dU}{dx} = -\left(\frac{1}{2}\frac{Q_0^2}{\ell^3\varepsilon_0}(-1)d\right) = \frac{Q_0^2 d}{2\ell^3\varepsilon_0}$$

Hence $\frac{F}{A} = \frac{F}{\ell d} = \frac{Q_0^2}{2\ell^4\varepsilon_0}$

Energy density $= \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 \frac{Q_0^2}{\left(\ell^2\varepsilon_0\right)^2} = \frac{Q_0^2}{2\ell^4\varepsilon_0}$

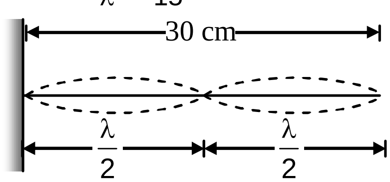
9. BCD

Sol. $y = 4\sin\left(\frac{\pi x}{15}\right)\cos(96\pi t)$ ....(i)

But $\quad y = 2a\sin kx \cos\omega t$

(A) $y_{max} = 4\sin\left(\frac{\pi}{15} \times 5\right) = 2\sqrt{3}\text{cm}$

(B) $k = \frac{2\pi}{\lambda} = \frac{\pi}{15} \Rightarrow \lambda = 30\text{cm}$



Thus, third node is formed at 30 cm from one end of the string.

(C) Differentiating eqn. (i) w.r.t. time:

$v = -4(96\pi)\sin\left(\frac{\pi x}{15}\right)\sin(96\pi t)$

At $x = 7.5$ and $t = 0.25$,

$v = -4(96\pi)\left(\sin\frac{\pi}{2}\right)\sin(24\pi) = 0$

(D) $y = y_1 + y_2$

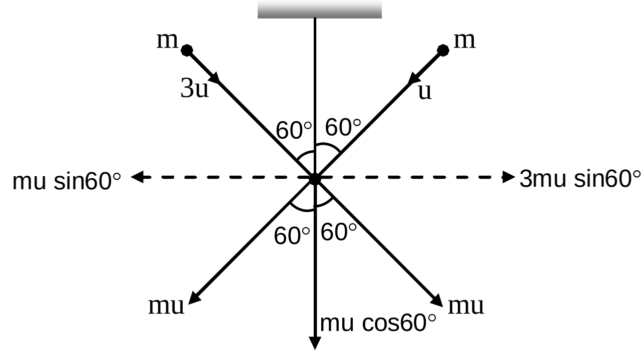
$y = 2\sin\left(\frac{\pi x}{15} - 96\pi t\right) + 2\sin\left(\frac{\pi x}{15} + 96\pi t\right)$

$y = 4\sin\frac{\pi x}{15}\cos 96\pi t$

Hence, options (B), (C) and (D) are correct.

10. AB

Sol. (i) Using the law of conservation of momentum along horizontal direction.



$$3\text{ mu}\sin 60° - \text{mu }\sin 60° = (10m + 2m)V$$

$$2\text{ mu }\sin 60° = 12mV$$

$$2\text{ mu}\left(\frac{\sqrt{3}}{2}\right) = 12mV$$

$$V = \frac{u\sqrt{3}}{12}$$

(ii) Impulse in the string = change in momentum in vertical direction

$$= \text{mu }\cos 60° + 3\text{mu}\cos 60° = 2\text{mu}$$

(iii) Initial K.E. $= \frac{1}{2}mu^2 + \frac{1}{2}m(3u)^2 = 5mu^2$

K.E. gained by the system $= \frac{1}{2}(12m)V^2 = \frac{mu^2}{8}$

$\therefore$ Loss in energy $= 5mu^2 - \frac{mu^2}{8} = \frac{39}{8}mu^2$

Section – B

11. 3

Sol. $\lambda_{th} = \frac{1240}{2} = 620$

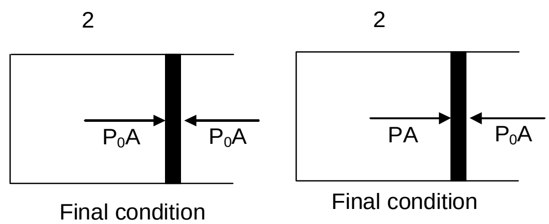
$$i = \frac{dN}{dt}e = \frac{e \times A}{hc}\int \lambda dt = \frac{eAk\int_{100}^{620}\lambda d\lambda}{hc}$$

$$= \frac{eAk\lambda^2}{2\lambda c}\Bigg|_{400}^{620} = \frac{1\times 1\times\left(620^2 - 400^2\right)\times k}{1240\times 2}$$

$$= \frac{220\times 1020}{1240\times 2}\times\frac{0.031}{100} = 2.8A \approx 3A$$

12. 7

Sol.



Volume of the cylinder remains constant $\frac{P_0}{T_0} = \frac{P}{2T_0}$

$\Rightarrow \quad P = 2P_0$

Now, $PA = P_0A + f$

$\Rightarrow \; (P - P_0)A = f$

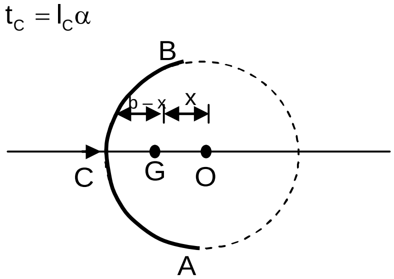
$\Rightarrow (2P_0 - P_0)A = \int \mu \frac{dN}{dl} dl = \frac{dN}{dl} \times \mu \times 2\pi r$

$\Rightarrow \frac{dN}{dl} = \frac{P_0 \times \pi r^2}{\mu \times 2\pi \times r} = \frac{P_0 r}{2\mu} = \frac{10^5 \times 5 \times 10^{-2}}{2 \times 0.2}$

$= 125 \times 10^2 \text{ N/m}$

13. 2

Sol. $t_C = I_C \alpha$



$\alpha = \frac{mg(b - x)}{I_C} = \frac{g}{2b}$

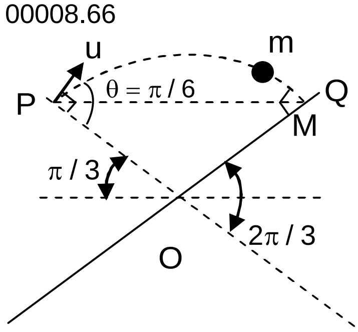
Using parallel axis theorem

$I_C = m2b(b - x)$

Section – C

14. 00008.66

Sol.



$t_{PQ} = \frac{T}{2} = 1\text{sec}$

$t_{PQ} = \frac{(2u\sin\theta)}{g}$

$\frac{2u\left(\frac{1}{2}\right)}{10} = 1$

U = 10 m/s

At highest point velocity of ball = $u\cos\theta = 5\sqrt{3}\ m/s$

15. 00017.32

Sol. Since POQ is equilateral triangle

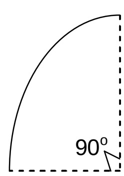
$\frac{\ell}{2} =$ Range of ball = PQ=$\frac{u^2\sin 2\theta}{g}$

$= 10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3}$

$\Rightarrow L = 10\sqrt{3}$ m

16. 00000.50

Sol. $R = 10m = \frac{mv}{qB}$



$\Rightarrow B = \frac{0.1\times 50}{1\times 10} = 0.5T$

17. 00003.00

Sol. $1.8 = \frac{1}{2}gt^2$

t = 0.6 sec

$\theta = \omega t = \frac{qB}{m}t = \frac{1}{0.1}\times 0.5\times 0.6 = 3$ rad

18. 00001.76

Sol. $h_0 = \sqrt{h_1h_2} = \sqrt{1.6\times 0.9} = 1.2mm = 2a\delta$

$1.2 = 2aA(\mu - 1)$

$\Rightarrow \mu \approx 1.76$

19. 00000.70

Sol. $d = 1.2$ mm; $D = 140$ cm

$\therefore \beta = \frac{\lambda D}{d} = 0.7$ mm

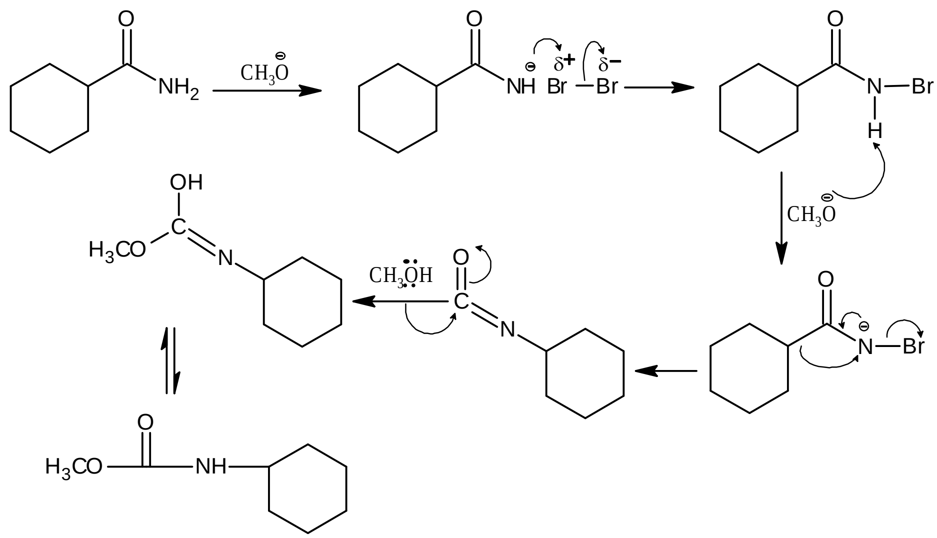
Chemistry

PART – II

Section – A

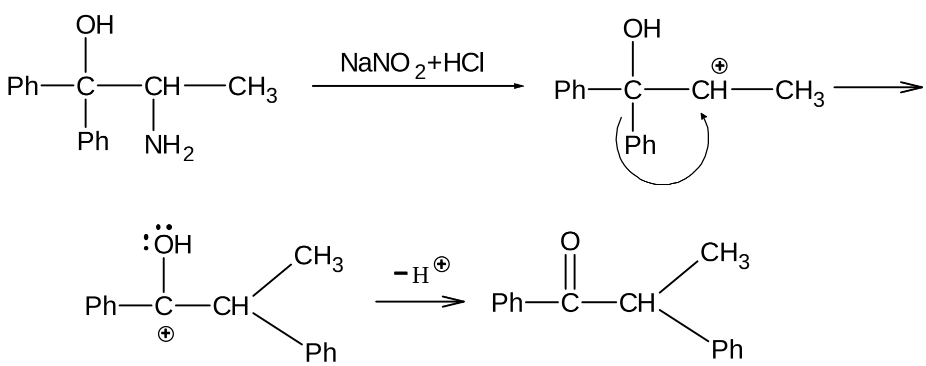
20. B

Sol.



21. C

Sol.



22. C

Sol. $A: Cl^- \longrightarrow \underset{P_1}{\frac{1}{2}Cl_2} + e^-$

$C: \underset{P_2}{\frac{1}{2}Cl_2} + e^- \longrightarrow Cl^- \quad \Rightarrow \qquad Q = \left[\frac{P_1}{P_2}\right]^{1/2}$

$$E_{cell} = 0 - \frac{0.059}{1}\log Q$$

For $E_{cell} > 0; P_1 < P_2$

23. B

Sol. For system consisting of two immiscible liquids.

$P_{Solution} = P^0_{H_2O} + P^0_{n-butylchloride} = 1\,atm = 760\,mm$

$P^0_{n-butyl\ chloride} = 760 - 570 = 190\,mm$ of Hg

Also in vapour state

$$\frac{n_{(H_2O)}}{n_{(n-\text{butyl chloride})}} = \frac{P^0_{H_2O}}{P^0_{n-\text{butyl chloride}}} = \frac{570}{190}$$

$$\frac{\omega_{H_2O}/18}{\omega_{n-\text{butyl chloride}}/92.5} = 3$$

$$\frac{\omega H_2O}{\omega_{n-\text{butyl chloride}}} = \frac{3\times 18}{92.5} = \frac{0.58}{1}$$

24. C

Sol. Option ABC are optically active because of chirality in molecule
Option D having two similar gp. so achiral molecule.

25. BCD

Sol. Based on nomenclature system of geometrical isomers.

26. ABD

Sol. By application of Jahn Teller distortion effect. In d^1, d^2, d^4, d^5 low spin d^6 high spin J. Teller effect observed.

27. AD

Sol. Red bauxite $\left(Al_2O_3\right) + KOH \longrightarrow KAlO_2 + H_2O$
Residue contain Fe_2O_3 as impurity

28. AB

Sol. **Hall – Heroult's process**
Aluminum is liberated at the cathode and gets located at the bottom of tank
Oxygen liberated at anode combines with the carbon of the anode to produce carbon monoxide.

29. AB

Sol. $\underset{(aq)}{H^+} + \underset{(aq)}{OH^-} \longrightarrow H_2O(\ell)$

$$\Delta H^0 = -13.7\times 4.18\ \text{kJ/mole}$$

$$\frac{Mv}{1000} = \text{mole}$$

Moles of HCl $= \dfrac{200\times 0.862}{1000}$ mole
= 0.1724 mole of H^+

Moles of $Ba\left(OH\right)_2 = \dfrac{200\times 0.431}{1000}$ mole
=0.0862 mole Ba $(OH)_2$
= 0.1724 mole of OH^-

Thus, $\left[H^+\right] = \left[OH^-\right]$, hence no reactant is left unreacted.

$\Delta_r H^0$ for 0.1724 mol H^+ or OH^- ion

$$= -0.1724\times 57.2 = -9.86\ \text{kJ}$$

Thus, (A) is correct.

$\Delta H_{calorimeter} = C_p \ \Delta T = 435\,JK^{-1}\,\Delta T\ K$

$= 435\Delta TJ = 0.435\Delta T\ kJ$

$\Delta H_{solution} = m.s\,\Delta T = 400 \times 4.184\,\Delta TJ$

$= 1673.6\Delta T\ J = 1.6736\Delta T\,kJ$

$\Delta H_{Total} = (0.435\Delta T + 1.6736\Delta T)kJ$

$\therefore \quad 9.86 = [0.435 + 1.6736]\Delta T$

$\Delta T = 4.676^0$

$(T_2 - T_1) = 4.676^0$

$\therefore \quad T_2 = T_1 + 4.676 = 20.48 + 4.676$

$= 25.16\,^oC$

Thus, (B) is correct.

Section – B

30. 7

Sol. $2Cu^{2+} + 4I^- \longrightarrow I_2$

$2I^- + \underset{\text{Conc.}}{H_2SO_4} + 2H^+ \longrightarrow I_2 + SO_2 \uparrow + 2H_2O$

$2I^- \xrightarrow{MnO_4^-/H^+} I_2$

$2I^- \xrightarrow{Cr_2O_7^{2-}/H^+} I_2$

$2I^- + Cl_2 \longrightarrow 2Cl^- + I_2$

$2I^- + CaOCl_2 + 2H^+ \longrightarrow CaCl_2 + H_2O + I_2$

$2I^- + 4H^+ + 2NO_2^- \longrightarrow I_2 + 2NO + 2H_2O$

(I^- converts into I_2 when it reacts with oxidizing agents)

31. 3

Sol. O.N. of Cr in X($FeO.Cr_2O_3$) = + 3

O.N. of Cr in Z $(Na_2Cr_2O_7) = +6$

32. 6

Sol. Isoelectric pH is the pH at which there is no migration of Zwitter ion on passing electric current.

$$pH = \frac{pH(\text{with NaOH}) + pH(\text{with HCl})}{2}$$

$$= \frac{9.714 + 2.286}{6} = 6$$

Section – C

33. 00019.20

Sol. $H_2O(\ell) \rightleftharpoons H^+ + OH^-$

$K_{eq} = K_w = [H^+][OH^-] = 1 \times 10^{-14}$ at 298 K

$\therefore \quad \Delta G^0 = -2.303\ RT\ \log K_w$

$= -2.303 \times 0.002 \times 298 \log 1 \times 10^{-14}$

$= 19.2\ kcal\ mol^{-1}$

34. 00018.46

Sol. $H^+ + OH^- \rightleftharpoons H_2O$

$\Delta G^0 = -19.2 \text{ kcal mol}^{-1}$

$\Delta H^0 = -13.7 \text{kcal mol}^{-1}$(standard value)

$\therefore \quad \Delta G^0 = \Delta H^0 - T\Delta S^0$

$$\therefore \quad \Delta S^0 = \frac{\Delta H^0 - \Delta G^0}{T}$$

$$\therefore \quad = \frac{-13.7 + 19.2}{298}$$

$$= 0.01846 \text{ kcal mol}^{1-}\text{K}^{-1}$$

35. 00004.06_00004.07

Sol. Mole fraction of CH_3OH = 0.1
Mole fraction of CH_2OH = 0.3
Mole fraction of H_2O = 0.6
Sum of mole fraction = 1.0
Thus, moles of $CH_3OH = 1(= 32\,g)$
Moles of $CH_3CH_2OH = 3(= 138\,g)$
Moles of $H_2O = 6(= 108\,g)$
Molality of CH_3OH

$$= \frac{\text{Moles of } CH_3OH}{\text{kg of solvent } (H_2O + CH_3CH_2OH)}$$

$$= \frac{1}{246/1000} = 4.065 \text{ molal}$$

36. 00049.64

Sol. Mass percentage of CH_3CH_2OH

$$= \frac{138}{278} \times 100 = 49.64\%$$

37. 00303.00

Sol. Cu in 15 g brass $= 15 \times \dfrac{90}{100} = 13.5\,g$

$$= \frac{13.5}{63.5}\text{mol} = 0.2126\,\text{mol}$$

HNO_3 required = 4 times of Cu taken

$$= 0.2126 \times 4 \text{ mol} = 0.8504 \text{ mol}$$

Zn in 15 g brass $= 1.5\,g = \dfrac{1.5}{65.0}\text{mol}$

HNO_3 required = 2.5 times of Zn taken

$$= \frac{1.5}{65.0} \times 2.5 = 0.0577 \text{ mol}$$

Total HNO_3 = 0.9081mol

Sinc, $\dfrac{\text{volume(mL)} \times \text{molarity}}{1000} = \text{mol}$

$$\therefore \quad V(mL) = \frac{0.9081 \times 1000}{3.0} = 303\,mL$$

38. 00009.52

Sol. NO_2 is due to Cu only.

Thus, NO_2 formed = 2 times of Cu taken

$= 2 \times 0.2126$ mol

$\therefore$ Volume at STP $2 \times 0.2126 \times 22.4L = 9.52L$

Mathematics

PART – III

Section – A

39. D

Sol. $S_1 + S_2 = \frac{1}{2}\left[\sum_{k=1}^{\infty}\left\{\frac{1}{(6k-1)^2} + \frac{1}{(6k+1)^2}\right\}\right]$

$\Rightarrow 2(S_1 + S_2) + 1 = \frac{1}{1^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{11^2} + ...$ --------------- (1)

Let $\alpha = \frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + ...$ ----------------- (2)

$\frac{\alpha}{2^2} = \frac{1}{2^2} + \frac{1}{4^2} + ...$ ------------------ (3)

$\frac{\alpha}{3^2} = \frac{1}{3^2} + \frac{1}{6^2} + ...$ ------------------ (4)

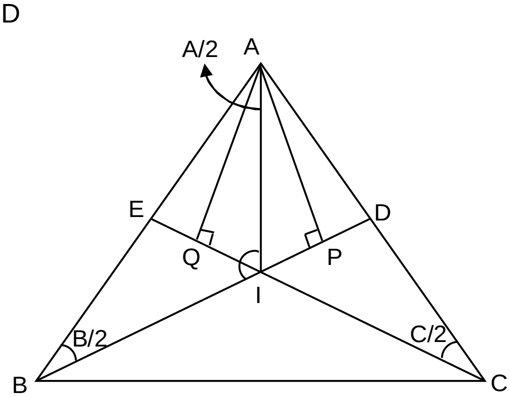
$\frac{\alpha}{6^2} = \frac{1}{6^2} + ...$ ------------------ (5)

Now, (2) – { (4) – (5) + (3)} gives

$\frac{2\alpha}{3} = 2(S_1 + S_2) + 1 \Rightarrow S_1 + S_2 = \frac{1}{2}\left[\frac{2\alpha}{3} - 1\right] = \frac{\pi^2}{18} - \frac{1}{2}$

40. D

Sol.



$\because \angle B + \angle C = \frac{2\pi}{3}, \ \therefore \angle A = \frac{\pi}{3}$

$\sin\frac{B}{2} = \frac{AP}{AB}$ and $\frac{BI}{\sin(A/2)} = \frac{AB}{\cos(C/2)}$

$\Rightarrow \frac{AP}{BI} = \frac{\sin(B/2)\cos(C/2)}{\sin(A/2)}$

similarly $\frac{AQ}{CI} = \frac{\sin(C/2)\cos(B/2)}{\sin(A/2)}$

$\Rightarrow \frac{AP}{BI} + \frac{AQ}{CI} = \cot\frac{A}{2} = \cot\frac{\pi}{6} = \sqrt{3}$.

41. B

Sol. Since, x, y, z are not all zero, therefore given system of equation has non-trivial solution

$$\therefore \quad \begin{vmatrix} 1 & -a_3 & -a_2 \\ a_3 & -1 & a_1 \\ a_2 & a_1 & -1 \end{vmatrix} = 0$$

$$\Rightarrow \quad a_1^2 + a_2^2 + a_3^2 + 2a_1a_2a_3 = 1 \qquad ...(1)$$

Since, $a_1 = m - [m]$ and m is not an integer.

$$\therefore \quad 0 < a_1 < 1 \Rightarrow 0 < 1 - a_1^2 < 1 \qquad ...(2)$$

From eq.(1),

$$1 - a_2^2 - a_3^2 = a_1^2 + 2a_1a_2a_3 \quad \Rightarrow \quad \left(1 - a_2^{\,2}\right)\left(1 - a_3^{\,2}\right) = \left(a_1 + a_2a_3\right)^2 \qquad ...(3)$$

Similarly,

$$\left(1 - a_1^{\,2}\right)\left(1 - a_3^{\,2}\right) = \left(a_2 + a_1a_3\right)^2 \qquad ...(4)$$

$$\left(1 - a_1^2\right)\left(1 - a_2^{\,2}\right) = \left(a_3 + a_1a_2\right)^2 \qquad ...(5)$$

From Eq. (5),

$$1 - a_2^{\,2} = \frac{\left(a_3 + a_1a_2\right)^2}{1 - a_1^{\,2}} > 0$$

From Eq. (4),

$$1 - a_3^{\,2} > 0 \Rightarrow 3 - \left(a_1^{\,2} + a_2^{\,2} + a_3^{\,2}\right) > 0 \Rightarrow a_1^{\,2} + a_2^{\,2} + a_3^{\,2} < 3 \Rightarrow 1 - 2a_1a_2a_3 < 3 \text{ [FromEq. (1]}$$

$$\Rightarrow \quad a_1a_2a_3 > -1$$

42. B

Sol. Let α be the common root, then

$\alpha^2 + a\alpha + [m+1] = 0$ ...(i)

$\alpha^2 + \alpha b + [m+4] = 0$ ...(ii)

$\alpha^2 - c\alpha + [m+15] = 0$ ...(iii)

adding Equations (i) and (ii) and subtracting Equation (iii)

$\alpha^2 + [m] - 10 = 0$ ...(iv)

but $m = \lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{r=1}^{2n} \frac{r}{\sqrt{n^2 + r^2}} = \int_0^2 \frac{x}{\sqrt{1+x^2}}dx = \left(\sqrt{1+x^2}\right)_0^2 = \sqrt{5} - 1$, Now, $[m] = \left[\sqrt{5} - 1\right] = 1$

Now, from equation (iv), $\alpha^2 + 1 - 10 = 0 \Rightarrow \alpha = \pm 3$ ...(v)

Now, number of determinants of order 2 having 0, 1, 2, 3 = 4! = 24.

Let $\Delta_1 = \begin{vmatrix} a_1 & a_2 \\ a_3 & a_4 \end{vmatrix}$ be one such determinant and there exists another determinant.

$\Delta_2 = \begin{vmatrix} a_3 & a_4 \\ a_1 & a_2 \end{vmatrix}$ (obtained on interchanging R_1 and R_2) such that $\Delta_1 + \Delta_2 = 0$

P = sum of all the 24 determinants = 0

Since, $\alpha > p \Rightarrow \alpha > 0$, From equation (v), $\alpha = 3$

43. A

Sol. Let A = B, then 2A + C = 180° and 2 tan A + tan C = 100

Now, 2 A + C = 180° $\Rightarrow$ tan 2A = –tan C ...(i)

Also, 2 tan A + tan C = 100 ...(ii)

$\Rightarrow$ 2 tan A – 100 = – tan C

From Equations (i) and (ii), we have

$$2\tan A - 100 = \frac{2\tan A}{1-\tan^2 A}$$

Let $\tan A = x$, then $\frac{2x}{1-x^2} = 2x - 100$

$\Rightarrow$ $x^3 - 50x^2 + 50 = 0$

Let $f(x) = x^3 - 50x^2 + 50 = 0$

Then, $f'(x) = 3x^2 - 100x$

Then, $f'(x) = 0$ has roots $0, \frac{100}{3}$

Also, $f(0)\cdot f\left(\frac{100}{3}\right) < 0$

Thus, f(x) = 0 has exactly three distinct real root. Therefore, tan A and hence A has three distinct values. Thus, there exist exactly three non-similar isosceles triangles

44. AC

Sol. $(ay - bx)^2 + 4xy = 0$

$\Rightarrow a^2y^2 + b^2x^2 + (4 - 2ab)xy = 0$

$\Rightarrow \frac{a^2y}{x} + \frac{b^2x}{y} - (2ab - a) = 0$

Put $\frac{y}{x} = t$

$\Rightarrow a^2t^2 - (2ab - 4)t + b^2 = 0$

$D = (2ab - 4)^2 - 4a^2b^2$

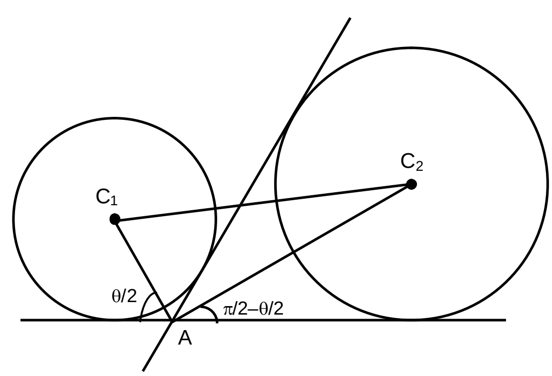
$= 4(4 - 4ab) = 16(1 - ab)$ must be a perfect square.

45. AB

Sol. From the figure it is clear that $\angle C_1AC_2 = 90^\circ$

Similarly $\angle C_1BC_2 = \angle C_1CC_2 = \angle C_1DC_2 = 90^\circ$

Thus ABCD is a cyclic quadrilateral with C_1C_2 as diameter ABCD is clearly not a square.



46. ABCD

Sol. (A) $\frac{a^2}{1} + \frac{b^2}{1} \ge \frac{(a+b)^2}{2}$

$\Rightarrow \frac{a^2+b^2}{a+b} \ge \frac{a+b}{2}$

$\Rightarrow \sum \frac{a^2+b^2}{a+b} \ge a+b+c = 8$

(C) as $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \ge \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca}$

$a^2b^2 + b^2c^2 + c^2a^2 \ge abc(a + b + c)$

$\Rightarrow a^2b^2 + b^2c^2 + c^2a^2 \geq 8abc$
(D) $a^2 + b^2 + c^2 \geq ab + bc + ca$
$3(a^2 + b^2 + c^2) \geq (a + b + c)^2$
$= 64$

$\Rightarrow a^2 + b^2 + c^2 \geq \frac{64}{3}$

$\Rightarrow a^2 + b^2 + c^2 \geq 18$

47. AC
Sol. $\sin\beta + \sin\gamma + \sin\delta = 1 - \sin\alpha$
$\cos2\alpha + 1 - 2\sin^2\beta + 1 - 2\sin^2\gamma + 1 - 2\sin^2\delta \geq 10/3$

$\Rightarrow \cos2\alpha \geq \frac{1}{3} + 2\left(\sin^2\gamma + \sin^2\beta + \sin^2\delta\right)$

$\Rightarrow 1 - 2\sin^2\alpha \geq \frac{1}{3} + 2\frac{(\sin\gamma + \sin\beta + \sin\delta)^2}{3} = \frac{1 + 2(1 - \sin\alpha)^2}{3}$

$\Rightarrow 3 - 6\sin^2\alpha \geq 2\sin^2\alpha - 4\sin\alpha + 3.$

$\Rightarrow 8\sin^2\alpha - 4\sin\alpha \leq 0 \Rightarrow \alpha \in \left[0, \frac{\pi}{6}\right]$

$\therefore$ A, C.

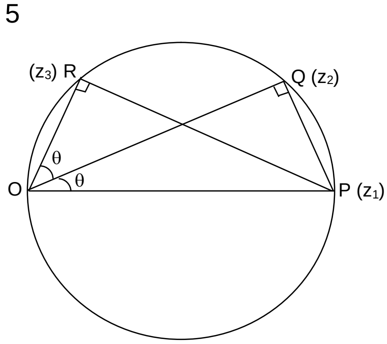
48. ABCD
Sol. Let f(x) = x

g(x) = $\frac{1}{x}$

f(x) g(x) = 1
periodic
(B) f(x) = x
g(x) = – x
f(x) + g(x) = 0
periodic
(C) f(x) = x
g(x) = x
f(x) – g(x) = 0
periodic
(D) f(x) = $\sin\sqrt{x}$
g(x) = x^2
f(g(x)) = sinx
Periodic.

Section – B

49. 5
Sol.



$\frac{z_2}{z_1} = \frac{OQ}{OP}e^{i\theta} \Rightarrow e^{i\theta} = \frac{z_2}{z_1 \cos\theta}$

$\frac{z_3}{z_1} = \frac{OR}{OP}e^{i2\theta} \Rightarrow e^{i2\theta} = \frac{z_3}{z_1 \cos 2\theta}$

$z_2^2 = z_1^2 \cos^2\theta \times \frac{z_3}{z_1 \cos 2\theta} \Rightarrow \frac{\cos^2\theta}{\cos 2\theta} = \frac{2+\sqrt{3}}{2\sqrt{3}}$

$\Rightarrow \cos^2\theta = \frac{2+\sqrt{3}}{4}$

$\Rightarrow 2\cos^2\theta = 1 + \cos 30° \Rightarrow \theta = 15^o$

50. 8

Sol. $x^2 + 81y^2 = 81$ ……. (1) and $x^2 + y^2 = 9$

$\Rightarrow 81x^2 + 81y^2 = 9 \times 81$ ……. (2)

Solving, (1) and (2), by subtracting

$80x^2 = 81 \times 8$

$\Rightarrow x^2 = \frac{81}{10}$, similarly, $y^2 = \frac{9}{10}$.

Above gives the point of intersection say (x_1, y_1).

Tangent at (x_1, y_1) to the circle $x^2 + y^2 = 9$ is

$xx_1 + yy_1 = 9$, $m_1 = -\frac{x_1}{y_1} = -\sqrt{9} = -3$.

Slope of tangent to ellipse is

$-\frac{x_1}{y_1} \times \frac{1}{81} = -\sqrt{\frac{9}{1}} \cdot \frac{1}{81} = m_2 \Rightarrow m_2 = -\frac{3}{81}$ $\therefore \tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{-3 + \frac{3}{81}}{1 + \frac{9}{81}}$

$\therefore \tan\theta = \frac{(-3 \times 80) \times 81}{81 \times 90} \Rightarrow \tan\theta = -\frac{8}{3} \Rightarrow -3\tan\theta = 8.$

51. 2

Sol. E = event of any one cutting a spade in one cut

$n(E) = {}^{13}C_1$

$n(S) = {}^{52}C_1$

$P(E) = 1/4 = p$, $P(\bar{E}) = q$

Probability of a winning = p + qqqqp + qqqqqqqqp+ ……∞

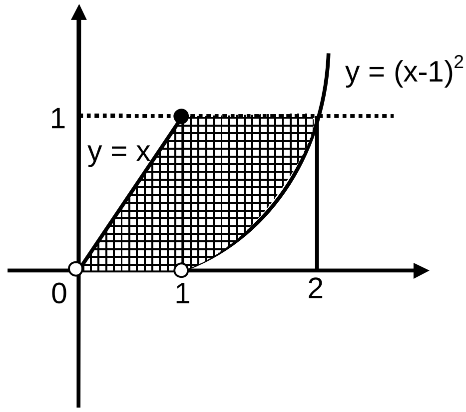
$= \frac{p}{1-q^4} = \frac{64}{175} = \lambda$

$\therefore 350\lambda = 350 \times \frac{64}{175} = 128 \Rightarrow \left[\frac{350\lambda}{64}\right] = 2.$

Section – C

52. 00001.00

Sol.



$0 \le (a-1)^2 < 1$

$\Rightarrow a \in [0, 2) \quad \Rightarrow \quad a = 0, 1$

53. 00001.17

Sol. Area will be bounded when $(a-1)^2 = 0$

$$A = \int_0^1 (x_2 - x_1)\,dy$$

and

$$\Rightarrow A = \int_0^1 \left(\left(\sqrt{y}+1\right) - y\right) dy = \frac{7}{6} = 1.17$$

54. 00000.52

Sol. Curve C is $(x-1)^2 + y^2 = 1$

and $x^2 - 3y^2 - 2x + 1 = 0$ represents

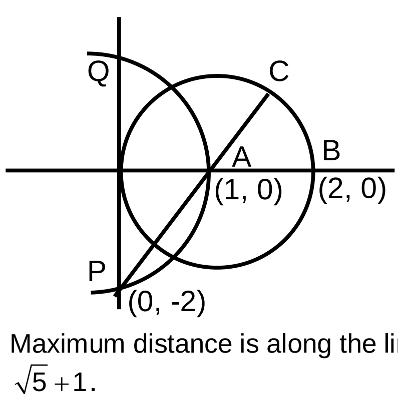
lines $x - \sqrt{3}\,y - 1 = 0$ and $x + \sqrt{3}\,y - 1 = 0$

passing through centre of circle (1, 0)

Hence area of smaller region $\frac{1}{2}r^2\theta = \frac{1}{2}\left(\frac{\pi}{3}\right) = \frac{\pi}{6}$

55. 00003.24

Sol.



Maximum distance is along the line through P passing from centre 'A' and is equal to $\sqrt{5}+1$.

56. 00001.00

Sol. We have $(t_2 + t_3)(t_2 + t_1) = -4$; $(t_4 + t_3)(t_4 + t_1) = -4$

Hence $t^2 + t(t_1 + t_3) + t_1t_3 + 4 = 0$ is satisfied by t_2 & t_4

So, $t_2 + t_4 = -1(t_1 + t_3)$ & $t_2t_4 = t_1t_3 + 4$

$\Rightarrow \left|\frac{t_2 + t_4}{t_1 + t_3}\right| = 1$

57. 00004.00

Sol. We have $(t_2 + t_3)(t_2 + t_1) = -4$; $(t_4 + t_3)(t_4 + t_1) = -4$

Hence $t^2 + t(t_1 + t_3) + t_1t_3 + 4 = 0$ is satisfied by t_2 & t_4

So, $t_2 + t_4 = -1(t_1 + t_3)$ & $t_2t_4 = t_1t_3 + 4$

$(t_1 - t_3)^2 = (t_1 + t_3)^2 - 4t_1t_3 = (t_2 + t_4)^2 - 4t_2t_4 + 16$

$= (t_2 - t_4)^2 + 16 \geq 16$

$\Rightarrow |t_1 - t_3|_{min} = 4$